GCSE CHEMISTRY

Higher Tier – Paper 2 style

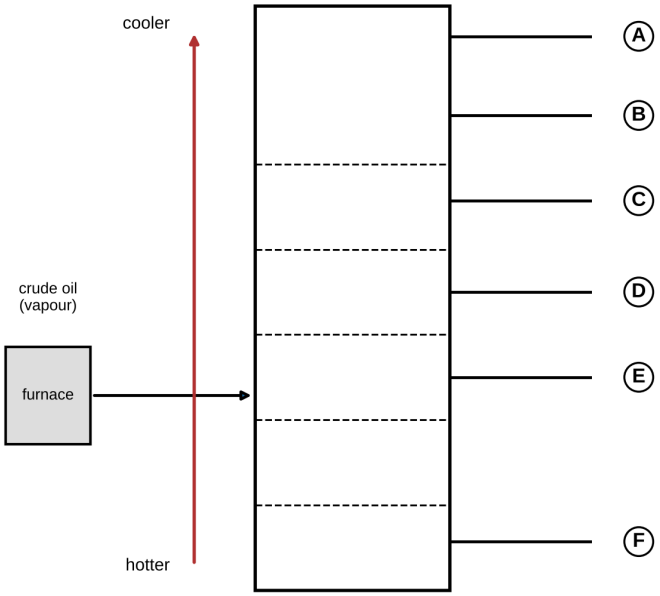
Topic 4.7: Organic chemistry

Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 130 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **126**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 126

0 1 Crude oil is separated into fractions by fractional distillation. The diagram shows a fractionating column. The letters **A** to **F** show the fractions collected.



0 1 . 1 What is meant by a **hydrocarbon**? **[1 mark]**

0 1 . 2 Crude oil is a mixture. Explain what is meant by a **mixture** in this context. **[2 marks]**

0 1 . 3 Describe how fractional distillation separates crude oil into fractions. **[4 marks]**

0 1 . 4 Fraction **A** is refinery gases and fraction **F** is bitumen. Which letter shows the fraction used as petrol? **[1 mark]**

0 1 . 5 Which fraction has the highest boiling point? Give a reason for your answer. **[2 marks]**

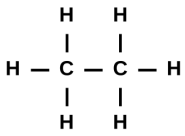
0 1 . 6 Compare fraction **A** and fraction **F**. Describe **two** differences in their properties. **[2 marks]**

0 1 . 7 Explain why the boiling point of a hydrocarbon increases as the length of the carbon chain increases. **[3 marks]**

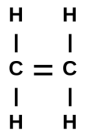
0 1 . 8 Give **one** use of fraction **F** (bitumen) and **one** use of kerosene. **[2 marks]**

0 2 The diagram shows the displayed formulae of four hydrocarbons, **A**, **B**, **C** and **D**.

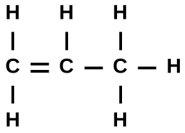
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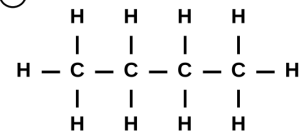
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0 2 . 1 Which **two** hydrocarbons are alkenes? **[1 mark]**

0 2 . 2 Give the name of hydrocarbon **D** and the name of hydrocarbon **C**. **[2 marks]**

0 2 . 3 Write the molecular formula of hydrocarbon **D**. **[1 mark]**

0 2 . 4 The general formula of the alkanes is C_nH_{2n+2}. **[1 mark]**
What is the molecular formula of an alkane with 8 carbon atoms?

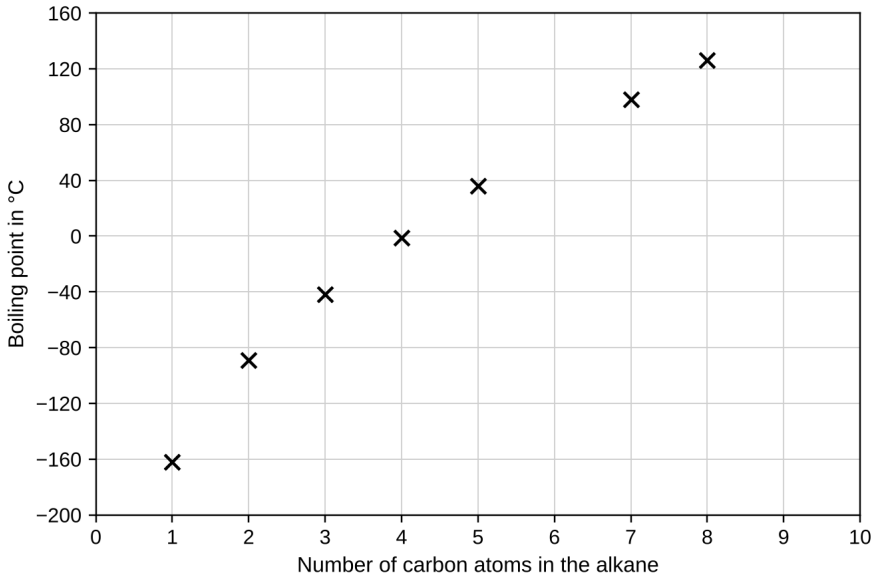
0 2 . 5 Alkanes are described as **saturated**. What does this mean? **[1 mark]**

0 2 . 6 Draw the displayed formula of propane. **[2 marks]**

0 2 . 7 Propane burns completely in oxygen. Write a balanced symbol equation for the reaction. **[2 marks]**

0 2 . 8 A gas cooker burns propane in a room with very little air. Describe what is produced when the propane burns incompletely, and explain why this is dangerous. **[3 marks]**

0 3 The graph shows the boiling points of some alkanes. The point for hexane (6 carbon atoms) is missing.



0 3 . 1 Draw a curve of best fit through the points and use it to estimate the boiling point of hexane. **[2 marks]**

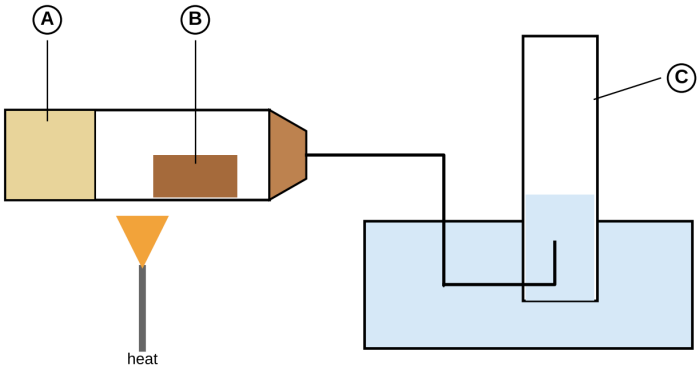
0 3 . 2 Describe the pattern shown by the graph. **[2 marks]**

0 3 . 3 Which alkanes in the graph are gases at room temperature (20 °C)? Give a reason. **[2 marks]**

0 3 . 4 Pentane is a liquid at 20 °C. Explain how the graph shows this. **[1 mark]**

0 3 . 5 A student says: 'As the carbon chain gets longer the alkane is easier to ignite.' Is the student correct? Explain your answer. **[2 marks]**

0 4 Some long-chain hydrocarbons are cracked to make shorter, more useful molecules. The diagram shows apparatus used in the laboratory to crack liquid paraffin.



0 4 . 1 What is in the parts labelled **A** and **B**? **[2 marks]**

0 4 . 2 Why does the demand for short-chain hydrocarbons exceed the amount produced by fractional distillation? **[2 marks]**

0 4 . 3 State **two** conditions needed for catalytic cracking. **[2 marks]**

0 4 . 4 Decane ($C_{10}H_{22}$) is cracked into octane and one other product. **[1 mark]**

$C_{10}H_{22} \rightarrow C_8H_{18} + ____$

0 4 . 5 Name the type of hydrocarbon that C_2H_4 is and give a test to show it is this type. **[3 marks]**

0 4 . 6 Cracking is described as thermal decomposition. Explain why. **[2 marks]**

0 4 . 7 Why is the gas collected over water in the diagram? **[1 mark]**

0 5 This question is about alkenes.

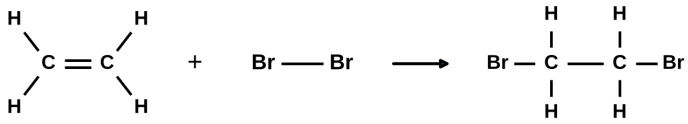
0 5 . 1 What is the functional group in alkenes? **[1 mark]**

0 5 . 2 The general formula of the alkenes is C_nH_{2n}. Give the molecular formula of pentene. **[1 mark]**

0 5 . 3 Alkenes are more reactive than alkanes. Explain why. **[1 mark]**

0 5 . 4 Ethene reacts with bromine in an addition reaction. The diagram shows the reaction. **[3 marks]**

Name the product, and explain why this is called an addition reaction.



0 5 . 5 A student shakes ethene with bromine water. Describe what the student sees. **[2 marks]**

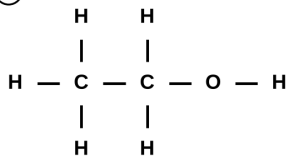
0 5 . 6 Alkenes burn with a smokier flame than alkanes. Explain why. **[2 marks]**

0 5 . 7 Ethene reacts with steam to make ethanol. **[2 marks]**

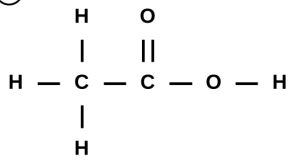
Write a balanced symbol equation for the reaction.

0 6 The diagram shows the displayed formulae of two organic compounds, **P** and **Q**.

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0 6 . 1 Name compound **P** and compound **Q**. **[2 marks]**

0 6 . 2 Name the functional group in **P** and the functional group in **Q**. **[2 marks]**

0 6 . 3 Ethanol is made by fermentation. Complete the word equation and state **two** conditions. **[3 marks]**

glucose → ethanol + ____

0 6 . 4 Ethanol burns completely in oxygen. Write a balanced symbol equation for the reaction. **[2 marks]**

0 6 . 5 Sodium is added to ethanol. Describe **one** thing you would see and name the gas produced. **[2 marks]**

0 6 . 6 Ethanoic acid is a weak acid. Explain what this means. **[1 mark]**

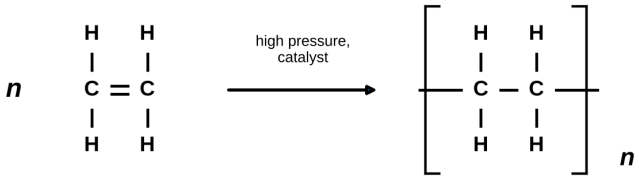
0 6 . 7 Sodium carbonate is added to ethanoic acid. Complete the word equation. **[2 marks]**

ethanoic acid + sodium carbonate → _____ + _____ + _____

0 6 . 8 Ethanoic acid reacts with ethanol to make an ester. Name the ester and give one use of esters. **[2 marks]**

0 6 . 9 Name the process that changes ethanol into ethanoic acid when wine is left open to the air. **[1 mark]**

0 7 Poly(ethene) is an addition polymer. The diagram shows how it is made from ethene.



0 7 . 1 Name the monomer used to make poly(ethene). **[1 mark]**

0 7 . 2 Explain why ethene molecules can join together to form a polymer. **[2 marks]**

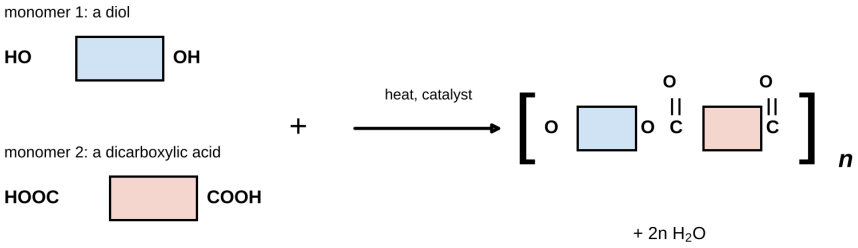
0 7 . 3 Propene is $CH_3CH=CH_2$. Draw the repeat unit of poly(propene) as a displayed formula. **[2 marks]**

0 7 . 4 Chloroethene (C_2H_3Cl) is the monomer for another plastic. Name the polymer. **[1 mark]**

0 7 . 5 Most addition polymers are not biodegradable. Explain why this causes a problem for waste disposal. **[3 marks]**

0 7 . 6 Explain why an addition polymerisation produces only one product. **[1 mark]**

0 8 Polyesters are made by condensation polymerisation. The diagram shows the monomers and the polymer.



0 8 . 1 Name the type of compound used as monomer 1 and monomer 2. **[2 marks]**

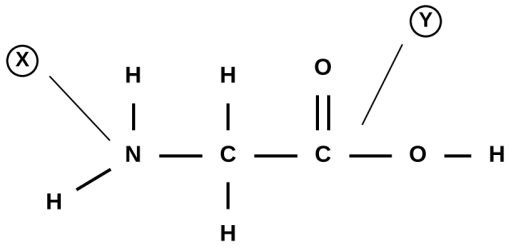
0 8 . 2 Name the small molecule produced when each link forms. **[1 mark]**

0 8 . 3 Name the type of link formed between the monomers. **[1 mark]**

0 8 . 4 Explain why each monomer must have two functional groups. **[2 marks]**

0 8 . 5 State **two** differences between addition polymerisation and condensation polymerisation. **[2 marks]**

0 9 The diagram shows the structure of an amino acid, glycine.



0 9 . 1 Name the functional groups labelled **X** and **Y**. **[2 marks]**

0 9 . 2 Amino acids join together by condensation polymerisation. Name the type of molecule made and the small molecule released. **[2 marks]**

0 9 . 3 DNA is a natural polymer. What does DNA stand for? **[1 mark]**

0 9 . 4 Describe the structure of a DNA molecule. **[2 marks]**

0 9 . 5 Name **two** other natural polymers. **[2 marks]**

1 0 Complete the table of organic compounds.

1 0 . 1 Complete the table. **[6 marks]**

Name	**Molecular formula**	**Homologous series**
propane		alkane
	C_4H_8	alkene
ethanol		alcohol
	CH_3COOH	carboxylic acid
pentane		alkane
	C_3H_6	alkene

1 1 Stretch questions (grade 7–9).

1 1 . 1 *Describe how crude oil is separated into fractions. Explain why the fractions have different boiling points and different uses. **[6 marks]**

1 1 . 2 *A student has two colourless liquids. One is hexane (an alkane) and the other is hexene (an alkene). **[6 marks]**

Describe how the student could tell them apart. Explain the results and write an equation for any reaction that happens.

END OF QUESTIONS

The Periodic Table of the Elements (elements 1–56)

1 **H** 1																4 **He** 2
7 **Li** 3	9 **Be** 4											11 **B** 5	12 **C** 6	14 **N** 7	16 **O** 8	19 **F** 9	20 **Ne** 10
23 **Na** 11	24 **Mg** 12											27 **Al** 13	28 **Si** 14	31 **P** 15	32 **S** 16	35.5 **Cl** 17	40 **Ar** 18
39 **K** 19	40 **Ca** 20	45 **Sc** 21	48 **Ti** 22	51 **V** 23	52 **Cr** 24	55 **Mn** 25	56 **Fe** 26	59 **Co** 27	59 **Ni** 28	63.5 **Cu** 29	65 **Zn** 30	70 **Ga** 31	73 **Ge** 32	75 **As** 33	79 **Se** 34	80 **Br** 35	84 **Kr** 36
85.5 **Rb** 37	88 **Sr** 38	89 **Y** 39	91 **Zr** 40	93 **Nb** 41	96 **Mo** 42	[98] **Tc** 43	101 **Ru** 44	103 **Rh** 45	106 **Pd** 46	108 **Ag** 47	112 **Cd** 48	115 **In** 49	119 **Sn** 50	122 **Sb** 51	128 **Te** 52	127 **I** 53	131 **Xe** 54
133 **Cs** 55	137 **Ba** 56	57–71															

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

MARK SCHEME – Topic 4.7: Organic chemistry

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	a compound containing hydrogen and carbon only	Do not accept 'a mixture of hydrogen and carbon'.	1	AO1 4.7.1.1
01.2	crude oil contains many different hydrocarbons ; not chemically joined together / can be separated by physical methods (boiling points) ;		2	AO1 4.7.1.1
01.3	crude oil is heated / vaporised ; the vapours enter the column, which is hot at the bottom and cooler at the top ; hydrocarbons rise and condense at the level where the temperature is below their boiling point ; fractions with similar boiling points / chain lengths are drawn off at different heights ;		4	AO1 4.7.1.2
01.4	B		1	AO2 4.7.1.2
01.5	F ; it condenses at the bottom of the column where it is hottest ;		2	AO2 4.7.1.2
01.6	A has shorter chains / smaller molecules ; A is more flammable / less viscous / lower boiling point (any two differences) ;	Two correct differences for 2 marks.	2	AO2 4.7.1.2
01.7	longer molecules have stronger intermolecular forces ; more energy is needed to overcome the (intermolecular) forces ; covalent bonds within the molecules do not break ;	Do not accept 'covalent bonds break'. Boiling breaks intermolecular forces only.	3	AO2 4.7.1.2
01.8	bitumen: road surfaces / roofing ; kerosene: aircraft fuel / heating ;		2	AO1 4.7.1.2
02.1	B and C		1	AO2 4.7.2.1
02.2	D: butane ; C: propene ;		2	AO2 4.7.1.3
02.3	C_4H_{10}		1	AO2 4.7.1.3
02.4	C_8H_{18}		1	AO2 4.7.1.3
02.5	contain only single bonds / no C=C double bonds ; have the maximum number of hydrogen atoms ;	Either answer.	1	AO1 4.7.1.3
02.6	three carbon atoms in a chain joined by single bonds ; each carbon has four bonds (8 hydrogen atoms: 3, 2 and 3) ;		2	AO2 4.7.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.7	$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$; correct formulae and balanced ;		2	AO2 4.7.1.3
02.8	carbon monoxide (and / or carbon) and water ; carbon monoxide is toxic / poisonous ; it combines with haemoglobin in red blood cells and reduces the amount of oxygen carried ;	Allow soot / carbon for the first mark.	3	AO2 4.7.1.3
03.1	smooth curve drawn through the points ; 65 – 72 °C ;	Accept any value that is read correctly from the candidate's curve between 65 and 72.	2	AO3 4.7.1.2
03.2	as the number of carbon atoms increases the boiling point increases ; the increase gets smaller for longer chains / the curve flattens ;		2	AO3 4.7.1.2
03.3	methane, ethane, propane and butane ; their boiling points are below 20 °C ;		2	AO3 4.7.1.2
03.4	its boiling point (36 °C) is above 20 °C (and melting point is below 20 °C)		1	AO3 4.7.1.2
03.5	no ; longer chains are less flammable / harder to ignite (and more viscous) ;		2	AO3 4.7.1.2
04.1	A: mineral wool soaked in (liquid) paraffin ; B: (broken porcelain / aluminium oxide) catalyst ;		2	AO1 4.7.1.4
04.2	short-chain hydrocarbons make better fuels (petrol) / are in greater demand ; crude oil contains more long-chain hydrocarbons than are needed ;		2	AO2 4.7.1.4
04.3	a catalyst (e.g. zeolite / aluminium oxide) ; high temperature (about 600–700 °C) ;	Allow 'steam' for steam cracking.	2	AO1 4.7.1.4
04.4	C_2H_4		1	AO2 4.7.1.4
04.5	alkene (ethene) ; add / shake with bromine water ; orange (brown) to colourless ;		3	AO2 4.7.2.2
04.6	a compound breaks down into simpler substances / molecules ; using heat ;		2	AO1 4.7.1.4
04.7	to collect / measure the gas (displaces water) ; the gas is not very soluble in water		1	AO3 4.7.1.4
05.1	C=C double bond / carbon–carbon double bond		1	AO1 4.7.2.1
05.2	C_5H_{10}		1	AO2 4.7.2.1
05.3	alkenes have a double bond (C=C) ; alkanes have only single bonds		1	AO2 4.7.2.1
05.4	1,2-dibromoethane ; the double bond breaks / opens ; bromine atoms add on to the carbon atoms (only one product) ;		3	AO2 4.7.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	bromine water is decolourised ; orange / brown to colourless ;	Do not accept 'clear'. Colourless is needed.	2	AO1 4.7.2.2
05.6	alkenes have a higher percentage of carbon (than alkanes) ; so combustion is more likely to be incomplete / more carbon (soot) is made ;		2	AO3 4.7.2.2
05.7	$C_2H_4 + H_2O \rightarrow C_2H_5OH$; correct formulae and balanced ;	Allow CH_3CH_2OH.	2	AO2 4.7.2.2
06.1	P: ethanol ; Q: ethanoic acid ;		2	AO2 4.7.3.1
06.2	P: hydroxyl / –OH ; Q: carboxyl / –COOH ;		2	AO1 4.7.3.1
06.3	carbon dioxide ; yeast (enzymes) ; warm (30–40 °C) and no oxygen / anaerobic ;	Allow 'about 37 °C'.	3	AO1 4.7.3.1
06.4	$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$; correct formulae and balanced ;		2	AO2 4.7.3.1
06.5	fizzing / bubbles / sodium dissolves ; hydrogen ;		2	AO2 4.7.3.1
06.6	it is only partly ionised in water		1	AO1 4.7.3.2
06.7	sodium ethanoate ; water and carbon dioxide (both for the mark) ;		2	AO2 4.7.3.2
06.8	ethyl ethanoate ; perfumes / flavourings ;		2	AO1 4.7.3.2
06.9	oxidation		1	AO1 4.7.3.2
07.1	ethene		1	AO1 4.7.4.1
07.2	ethene has a (C=C) double bond ; the double bond opens so each monomer can join to the next ;		2	AO2 4.7.4.1
07.3	two carbon atoms in the chain joined by a single bond ; one carbon has a CH_3 group, the other has two H atoms; bonds extending through the brackets ;	Allow the repeat unit with n shown outside the brackets.	2	AO2 4.7.4.1
07.4	poly(chloroethene) / PVC		1	AO1 4.7.4.1
07.5	microorganisms cannot break them down ; they stay in landfill for a long time / take up land ; burning them can release toxic gases / carbon dioxide ;		3	AO3 4.7.4.1
07.6	all the atoms in the monomers end up in the polymer / nothing is lost		1	AO3 4.7.4.1
08.1	monomer 1: diol (alcohol with two –OH groups) ; monomer 2: dicarboxylic acid (two –COOH groups) ;		2	AO1 4.7.4.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.2	water		1	AO1 4.7.4.2
08.3	ester (link / bond)		1	AO1 4.7.4.2
08.4	so each monomer can react at both ends ; to form a long chain / polymer ;		2	AO2 4.7.4.2
08.5	addition: one monomer type with C=C, only the polymer is made ; condensation: two functional groups per monomer, a small molecule (water) is also made ;		2	AO1 4.7.4.2
09.1	X: amine / amino / NH_2 ; Y: carboxyl / carboxylic acid / COOH ;		2	AO1 4.7.4.3
09.2	polypeptide / protein ; water ;		2	AO1 4.7.4.3
09.3	deoxyribonucleic acid		1	AO1 4.7.4.3
09.4	made of nucleotides ; two polymer chains / strands in a double helix ;		2	AO1 4.7.4.3
09.5	starch ; cellulose / proteins / rubber (any other) ;	Allow 'carbohydrates'. Do not accept 'plastic'.	2	AO1 4.7.4.3
10.1	propane: C_3H_8 ; butene: C_4H_8 ; ethanol: C_2H_5OH ; ethanoic acid: CH_3COOH ; pentane: C_5H_{12} ; propene: C_3H_6 ;	One mark for each correct row.	6	AO1 4.7.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
11.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • crude oil is heated and the vapours enter a fractionating column that is hotter at the bottom and cooler at the top • hydrocarbons condense at different heights depending on their boiling points; fractions are drawn off • short-chain hydrocarbons have low boiling points and rise to the top; long chains condense near the bottom • boiling point increases with chain length because intermolecular forces get stronger and more energy is needed to separate molecules • short chains (gases, petrol) are flammable and runny so used as fuels; long chains (bitumen) are viscous, used for roads • heating oil / diesel / kerosene have intermediate chain lengths and properties for engines and heating		6	AO1 4.7.1.2
11.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • add bromine water to each liquid and shake • hexene turns bromine water from orange to colourless; hexane stays orange • an addition reaction happens because hexene has a C=C double bond; alkanes are saturated so do not react • equation: $C_6H_{12} + Br_2 \rightarrow C_6H_{12}Br_2$ • alternatively burn small samples: hexene gives a smokier, more sooty flame • because hexene has a higher percentage of carbon so burns with more incomplete combustion		6	AO3 4.7.2.2

Where marks are lost in Topic 4.7

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Boiling point:** when a hydrocarbon boils you break *intermolecular forces*, not covalent bonds. Longer chain = stronger forces = higher boiling point = more viscous = less flammable.
- ☐ **Fractional distillation:** hot at the bottom, cool at the top. Short chains rise to the top. Always say the fractions condense where the temperature is below their boiling point.
- ☐ **Alkanes vs alkenes:** alkanes C_nH_{2n+2} are saturated (single bonds). Alkenes C_nH_{2n} have C=C and are unsaturated. Test: bromine water goes from orange to *colourless* (not 'clear').
- ☐ **Combustion:** complete = carbon dioxide + water. Incomplete (limited oxygen) = carbon monoxide and/or carbon + water. Carbon monoxide is toxic because it stops haemoglobin carrying oxygen.
- ☐ **Cracking:** a long alkane breaks into a shorter alkane + an alkene. Check the carbons and hydrogens balance. Conditions: catalyst and high temperature, or steam.
- ☐ **Displayed formulae:** every carbon must have exactly four bonds. Draw all bonds including C–H and O–H. Check you have not left out a hydrogen.
- ☐ **Addition polymers:** the C=C opens and the repeat unit has two carbons in the chain. Draw the bonds sticking out through the brackets and add n. Condensation polymers lose a small molecule (water).
- ☐ **Functional groups:** alkene C=C, alcohol –OH, carboxylic acid –COOH, ester –COO–. Learn the name *and* the formula of each.
- ☐ **Fermentation:** sugar → ethanol + carbon dioxide using yeast at about 30–40 °C with no oxygen.